15.076 Project Proposal

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Problem Summary

Spain's Guadalquivir basin is the country's most agriculturally productive region, supplying over 800,000 hectares of irrigated farmland. However, it is in the grip of its most severe drought cycle in decades. Reservoir levels regularly fall below 30% capacity, yet water is still allocated according to rigid historical water rights that do not adapt to shrinking supply or changing crop conditions. We propose an optimization framework that determines how much water each major reservoir in the basin should release to each irrigation district each month, in order to maximize total agricultural yield subject to supply constraints, and to do so robustly under the deep uncertainty of future drought conditions.

Data

We have identified four open-access datasets:

- MITECO Boletín Hidrológico Semanal: Weekly water volume, capacity, and inflow records for every mainland Spanish reservoir larger than 5 hm^3, from 1988 to present. We will use the Guadalquivir reservoir inflow time series as training data for our LSTM and as the supply input to our LP. www.miteco.gob.es/es/agua/temas/evaluacion-de-los-recursos-hidricos/boletin-hidrologico.html
- ESYRCE (Encuesta sobre Superficies y Rendimientos de Cultivos): Annual irrigated crop area, yields, and irrigation method (drip, sprinkler, gravity) by province and crop type, from 1990 to 2023. We will use this to estimate each irrigation district's seasonal water demand as a function of its crop mix, forming the demand constraints of our optimization. mapa.gob.es/es/estadistica/temas/estadisticas-agrarias/agricultura/esyrce
- Spanish Drought Catalogue v1.0: A catalogue of 40 major drought episodes across mainland Spain from 1916 to 2020, with each event characterized by duration, spatial extent, and SPI severity at 10×10 km resolution. We will use these labeled episodes to construct the uncertainty set for our robust LP and to test our allocation policy against real historical drought scenarios. digital.csic.es/handle/10261/331384
- AEMET OpenData: Daily precipitation and temperature records from Spain's national meteorological agency. We will aggregate upstream catchment weather to weekly resolution and use it as input features to the LSTM alongside lagged reservoir levels. opendata.aemet.es/opendata/api

Methodology

1. LSTM inflow forecasting: We train an LSTM on MITECO reservoir time series to predict seasonal inflows one quarter ahead, outputting a forecast distribution that defines the uncertainty set for optimization.
2. Demand estimation: We fit regressions on ESYRCE crop data and climate covariates to estimate each district's seasonal water demand by crop mix.
3. Robust LP: We maximize total yield by deciding monthly hm^3 releases per reservoir-district pair, subject to capacity, ecological flows, and demand constraints; this is extended to a robust LP using the LSTM's uncertainty bounds.